

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

Mr. Rawson • IB DP Physics

# Simple Harmonic Motion 6 – Rotation and $\omega$

Where the SHM equations come from

## IB Physics Guide • §C.1 Simple harmonic motion • Theme C, Wave Behaviour

This worksheet will cover the following topics:

**C1-2** the defining equation  $a = -\omega^2 x$ **C1-3** describing an oscillator by period, frequency, angular frequency, amplitude, equilibrium position and displacement

**C1-4**  $T = \frac{1}{f} = \frac{2\pi}{\omega}$

**C1-9**  $x = x_0 \sin(\omega t + \phi)$ ,  $v = \omega x_0 \cos(\omega t + \phi)$ ,  $v = \pm \omega \sqrt{x_0^2 - x^2}$

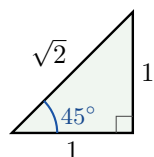
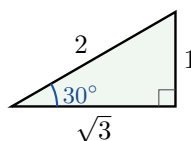
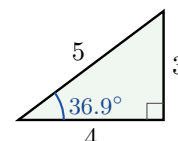
**Linking question**How can circular motion be used to visualize SHM? (*links to A.2*)

### Background Knowledge

*Read this as you come in. Highlight the key words, and use the space to add your own notes.*

#### A. Three triangles worth knowing by heart

Three right-angled triangles turn up so often that their sides are worth memorising. Each one gives you exact values of sin and cos with no calculator.

**Half a square.** Two equal sides, so the angles are  $45^\circ$  and  $45^\circ$ .  $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.71$ **Half an equilateral triangle.** Sides 1,  $\sqrt{3}$ , 2, angles  $30^\circ$  and  $60^\circ$ .  $\sin 30^\circ = \frac{1}{2}$ ,  $\cos 30^\circ = \frac{\sqrt{3}}{2} \approx 0.87$ **The 3–4–5 triangle.**  $3^2 + 4^2 = 5^2$ , so the sides are whole numbers.  $\sin 36.9^\circ = \frac{3}{5}$ ,  $\cos 36.9^\circ = \frac{4}{5}$ 

In every right-angled triangle:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

## B. Cos is the $x$ , sin is the $y$

Put a right-angled triangle inside a circle of radius 1, with one corner at the centre. The hypotenuse is the radius, so it is 1, and the two other sides are simply

$$x = \cos \theta \quad y = \sin \theta$$

**cos** measures how far across, **sin** how far up.

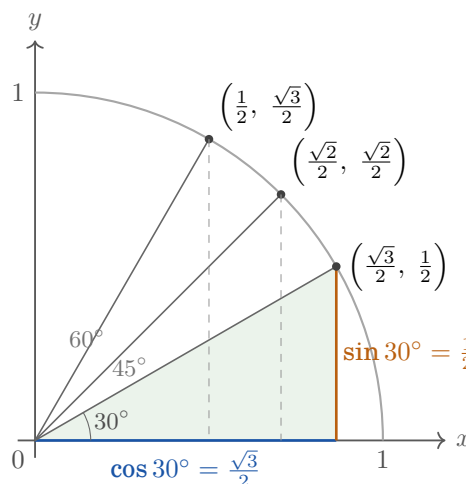
For a circle of radius  $r$ , every length scales by  $r$ :

$$x = r \cos \theta \quad y = r \sin \theta$$

**Example.**  $r = 5$  at  $36.9^\circ$  gives  $x = 5 \times \frac{4}{5} = 4$  and  $y = 5 \times \frac{3}{5} = 3$ : the 3-4-5 triangle again.

| angle $\theta$                                      | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$ |
|-----------------------------------------------------|-----------|----------------------|----------------------|----------------------|------------|
| <b>sin <math>\theta</math> (the <math>y</math>)</b> | 0         | $\frac{1}{2}$        | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | 1          |
| <b>cos <math>\theta</math> (the <math>x</math>)</b> | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$        | 0          |

Read along the rows: sin climbs from 0 to 1 as cos falls from 1 to 0. They are the same numbers in the opposite order, because  $\sin \theta = \cos(90^\circ - \theta)$ .



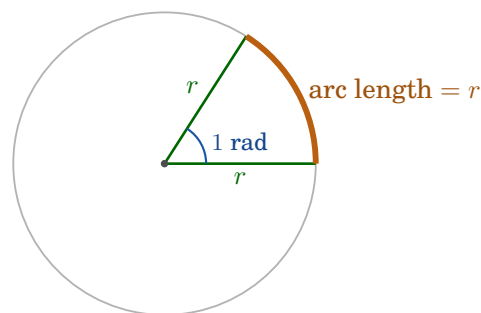
## C. Radians: measuring an angle in radii

Take the radius of a circle and bend it round the edge. The angle it covers is **one radian**.

$$\theta \text{ (in radians)} = \frac{\text{arc length } s}{\text{radius } r} \quad \text{so} \quad s = r\theta$$

The whole circumference is  $2\pi r$ , which is  $2\pi \approx 6.28$  radii laid end to end. So a full turn is  $2\pi$  radians, and

$$1 \text{ rad} = \frac{360^\circ}{2\pi} \approx 57.3^\circ$$



| degrees | $30^\circ$      | $45^\circ$      | $60^\circ$      | $90^\circ$      | $180^\circ$ | $360^\circ$ |
|---------|-----------------|-----------------|-----------------|-----------------|-------------|-------------|
| radians | $\frac{\pi}{6}$ | $\frac{\pi}{4}$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ | $\pi$       | $2\pi$      |

To convert, remember  $180^\circ = \pi \text{ rad}$ : multiply degrees by  $\frac{\pi}{180}$ , and radians by  $\frac{180}{\pi}$ . **Check:**

$$90^\circ \times \frac{\pi}{180} = \frac{\pi}{2} \approx 1.57.$$

**Today's question:** how does something going round a circle make an oscillation, and what does  $\omega$  measure?

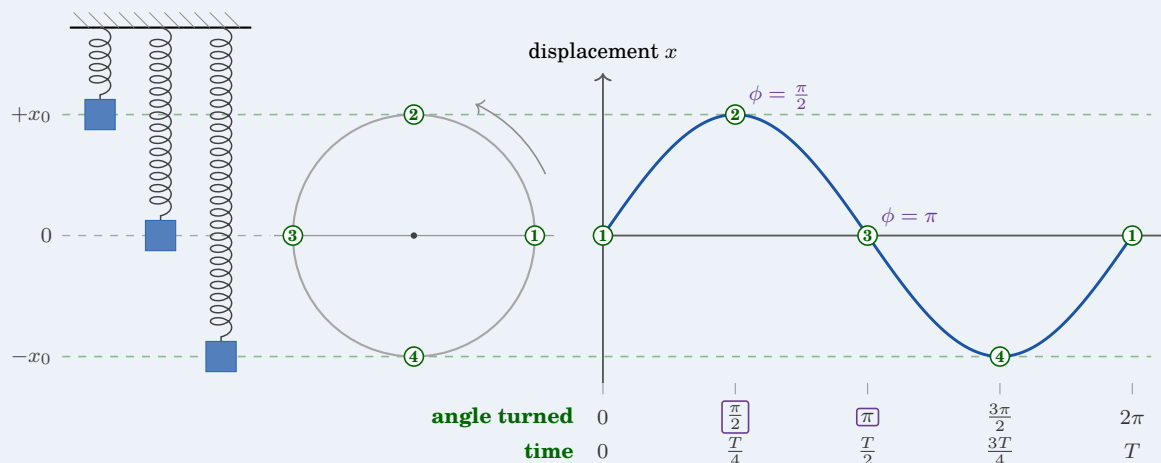
### Before you start

Set your calculator to **radians** and leave it there for the whole sheet. **Check:**  $\sin(\pi/2)$  gives exactly 1, and  $\sin(1.57)$  gives 1.000.

## Period, frequency and $\omega$

### One turn of the circle is one oscillation

Draw a circle whose radius is the amplitude  $x_0$ , beside the spring. As the point goes round, its height follows the mass exactly: up to  $+x_0$ , back through 0, down to  $-x_0$  and back again. The numbers show the same moment in all three pictures.



### One full turn

$= 2\pi$  radians = **one complete oscillation**, from ① round to ① again

### Period $T$ , in s

the time for one full turn, so the time for one oscillation

### Frequency $f$ , in Hz

oscillations per second:  $f = \frac{1}{T}$

### Angular frequency $\omega$ , in $\text{rad s}^{-1}$

radians turned per second. The point turns  $2\pi$  radians in a time  $T$ , so

$$\omega = \frac{2\pi}{T} = 2\pi f$$

### Phase angle $\phi$ (HL)

Starting position angle at time  $t$  is  $\omega t + \phi$ .

If the oscillation began at ① gives  $\phi = 0$

If the question started at ②  $\phi = \frac{\pi}{2}$

If the question started at ③  $\phi = \pi$

**Example.** A mass bounces with a period of 0.50 s. Then  $f = 1/0.50 = 2.0 \text{ Hz}$ , and  $\omega = 2\pi/0.50 = 4\pi \approx 12.6 \text{ rad/s}$ : the point goes round twice a second, turning 12.6 radians every second.

### Angular speed and angular frequency: two ideas, one symbol

|                  | Angular speed $\omega$                                                              | Angular frequency $\omega$                                                      |
|------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Describes</b> | something that is <b>actually turning</b> : a wheel, a record, a planet             | something that <b>repeats</b> : a spring, a pendulum. Nothing has to turn       |
| <b>Means</b>     | the angle the object turns through per second                                       | how many radians of the cycle pass per second, with $2\pi$ for each oscillation |
| <b>Equations</b> | $\omega = \frac{\Delta\theta}{\Delta t}$ $v = \omega r$ , so $\omega = \frac{v}{r}$ | $\omega = \frac{2\pi}{T} = 2\pi f$                                              |
| <b>Unit</b>      | rad/s                                                                               | rad/s                                                                           |
| <b>Topic</b>     | A.2 Circular motion                                                                 | C.1 Simple harmonic motion                                                      |

**Why they share a symbol.** An oscillation behaves like the shadow of a point going round a circle. The oscillation's **angular frequency** is the **angular speed** of that point.

#### Worked example — angular speed of a record

An LP turns at  $33\frac{1}{3}$  revolutions per minute.

$$f = \frac{33.33}{60} = 0.556 \text{ Hz} \quad T = \frac{1}{f} = 1.80 \text{ s} \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{1.80} = 3.49 \text{ rad/s}$$

In 0.50 s it turns through  $\Delta\theta = \omega \Delta t = 3.49 \times 0.50 = 1.75 \text{ rad}$ .

**Check:** 0.50 s is a little over a quarter of the period, and 1.75 rad is a little over a quarter turn ( $\pi/2 = 1.57$ ).

- 1) **Calculate** the angular speed of a bicycle wheel that makes one full turn every 0.40 s, and the angle it turns through in 0.10 s.

[3]

- 2) **Calculate** the angular speed of a single played at 45 revolutions per minute.

[2]

## From the circle to the SHM equations

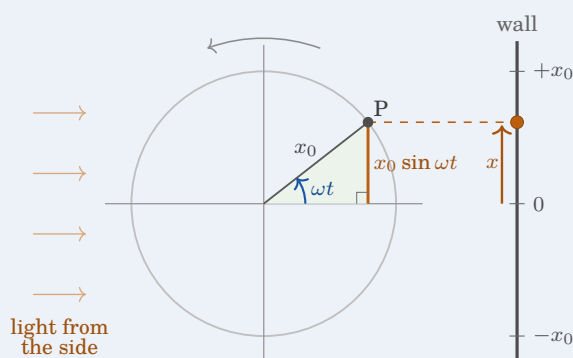
Each of the three equations below comes from the circle in a few steps.  $x = x_0 \sin \omega t$  and  $v = \omega x_0 \cos \omega t$  are **HL** (C1-9);  $a = -\omega^2 x$  is SL, for everyone (C1-2).

### 1. Displacement

1. From page 2, box B: a point on a circle is at height  $y = r \sin \theta$ .
2. The radius is the amplitude:  $r \rightarrow x_0$ .
3. The angle grows steadily,  $\omega$  radians every second:  $\theta \rightarrow \omega t$ .
4. The height of the point is the displacement:  $y \rightarrow x$ .

$$x = x_0 \sin \omega t$$

**Check:** at  $t = 0$ ,  $x = 0$ ; at  $\omega t = \pi/2$ ,  $x = +x_0$ . These are points ① and ② on page 3.



### 2. Velocity

1. Once round any circle is its circumference,  $2\pi r$ .
2. Here the radius is the amplitude,  $r = x_0$ , so once round is  $2\pi x_0$ .
3. The point goes once round in one period  $T$ , so

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{2\pi x_0}{T}$$

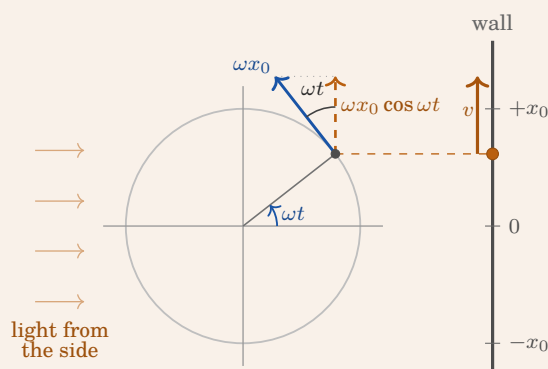
4. From page 3,  $\omega = \frac{2\pi}{T}$ , so

$$\text{speed} = \frac{2\pi}{T} x_0 = \omega x_0$$

5. It moves along the circle, at right angles to the radius, so its velocity makes the angle  $\omega t$  with the vertical.
6. The oscillation moves only up and down, so it takes the vertical part.

$$v = \omega x_0 \cos \omega t$$

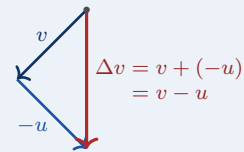
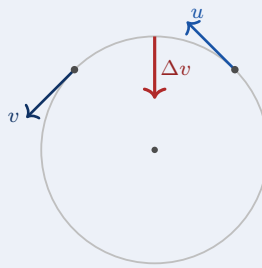
**Check:** at  $t = 0$ ,  $\cos 0 = 1$ , so it passes equilibrium at its fastest.



### Why the acceleration points at the centre

Acceleration is the change in velocity per second:

$$a = \frac{\Delta v}{\Delta t} \quad \Delta v = v - u$$



### 3. Acceleration

1. The velocity arrow, of length  $\omega x_0$ , also turns once every period. Its tip goes round a circle of radius  $\omega x_0$ , so, as in box 2, it changes at

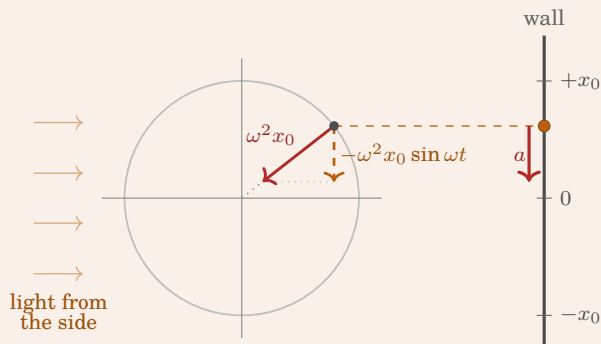
$$\frac{2\pi(\omega x_0)}{T} = \omega^2 x_0$$

pointing at the centre.

2. Take the vertical part. It points down while the point is above the centre, hence the minus sign:  $a = -\omega^2 x_0 \sin \omega t$ .
3.  $x_0 \sin \omega t$  is the displacement  $x$ .

$$a = -\omega^2 x$$

The defining equation of SHM.



### Given in your IB data booklet

You do not need to memorise these. You need to know where they come from, which is this sheet, and how to use them.

#### SL

$$T = \frac{1}{f} = \frac{2\pi}{\omega}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$a = -\omega^2 x$$

Basic circular motion equations

Spring-mass oscillators

Pendulum oscillators

The defining SHM relationship,  $a \propto -x$

#### HL

$$x = x_0 \sin(\omega t + \phi)$$

$$v = \omega x_0 \cos(\omega t + \phi)$$

$$v = \pm \omega \sqrt{x_0^2 - x^2}$$

$$E_T = \frac{1}{2} m \omega^2 x_0^2$$

$$E_p = \frac{1}{2} m \omega^2 x^2$$

Displacement at any time

Velocity at any time

Velocity at any displacement

Total energy of the oscillator

Potential energy at displacement  $x$